

provided for the software environment by Arduino and for the hardware environment by Intel at <http://www.intel.com/support/go/galileo>

Schools are using the boards to introduce students to embedded systems. Many students are fascinated at the prospect of using increased computing power to extend capabilities of existing projects. See a summary of the breadth of proposed applications at <http://dgit.in/UniversitiesGalileo> or check out the university maker community at <http://dgit.in/maker-scommunity> where universities will be posting their projects.

INTERNET OF THINGS EXPLORE CHALLENGE

Taking a lead in the Internet of Things (IoT) revolution, Intel has initiated the Intel® IoT Explorer Challenge, an IoT hackathon where selected participants will receive Intel® Galileo Development Kits to implement their ideas in the following categories: Home, Work, Play and Code for Good. Hobbyists, Students & Entrepreneurial Developers 21 years of age and older can check further details at <http://dgit.in/intelinternetofthings>

Intel India also hosts the Intel India Embedded Challenge, where selected finalists get kits based on Intel® architecture such as Intel® Atom™ or/ and Intel® Galileo and mentoring support (in case of students) for prototyping their project. The 2014 Embedded Challenge can be checked at <http://dgit.in/IntelEmbeddedChallenge>

BEYOND GALILEO

The Intel® Edison development platform is the first in a series of low-cost, general purpose compute platforms meant to cater to a wide variety of categories – from consumer electronics to industrial companies working on the Internet of Things (IoT).

Intel® Edison combines multiple features into its small size, delivering great performance, durability and a broad spectrum of I/O and software to help meet the needs of a wide range of customers.

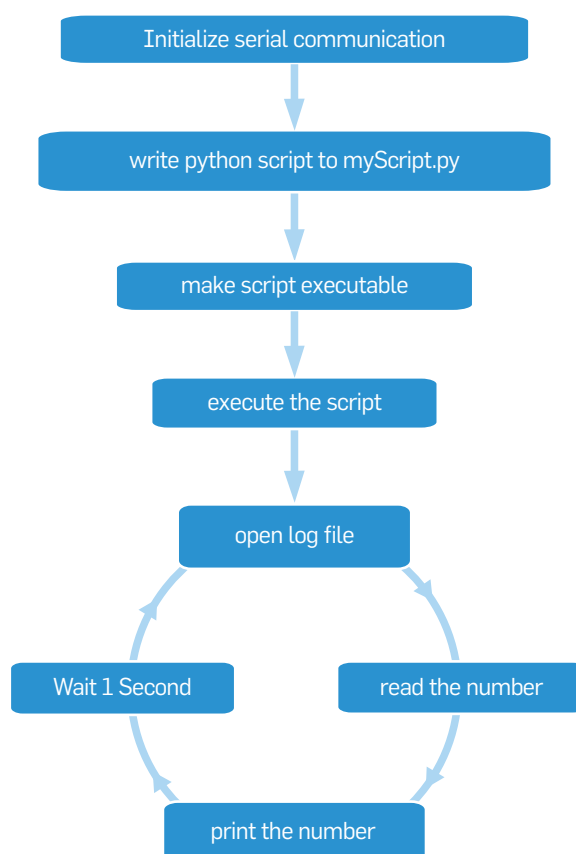
Those interested in the Galileo board can check the Intel site at <http://dgit.in/intelwheretobuy>. It has a vendor Mouser that sells the Galileo board in India.

You can get inspiration for Galileo-based projects at its community for demo projects at <http://dgit.in/inteldemoprojects>

Intel support team for Galileo boards can be accessed at <http://www.intel.com/support/galileo/emailsupport.htm>

Some of the great features of the Edison board are:

- Intel® Atom™ system-on-a-chip (SoC) based on 22nm fabrication Silvermont microarchitecture including a dual-core CPU and single core microcontroller
- Integrated Wi-Fi, Bluetooth LE, memory and storage
- Support for more than 30 industry-standard I/O interfaces via a 70-pin connector



Flowchart for executing Linux code on Galileo board

- Support for Yocto Linux, Arduino, Python, Node.js and Wolfram
- Open Source community software tools to enable easy adoption and help developers.

The unique combination of its small size with the power of the Edison board can empower developers to build the Internet of Things.

More info on the Edison board can be checked at <http://dgit.in/EdisonDevBoard>

To sum up, the Galileo board has wide applications in many areas, with a large number academic participants interested in the field of robotics and wearable electronics. Combining it with the Edison board can increase its capabilities manifold for seamless integration into myriad applications.

GALILEO HARDWARE SPECIFICATIONS PROCESSOR

- Instruction Set Architecture (ISA)-compatible 32-bit Intel® Pentium® processor
- 16 Kbytes L1 cache
- 512 Kbytes of on-die embedded SRAM
- Simple to program: single thread, single core and constant speed
- ACPI-compatible CPU sleep states supported
- Integrated real-time clock (RTC) with optional 3V 'coin cell' battery for operation between turn on cycles
- 400 MHz clock speed

STORAGE OPTIONS

- 8 Mbyte Legacy SPI Flash to store firmware (bootloader) and the latest sketch
- Between 256 Kbytes and 512 Kbytes dedicated for sketch storage
- 512 Kbytes embedded SRAM
- 256 Mbytes DRAM
- Optional micro SD card offers up to 32 Gbytes of storage
- USB storage works with any USB 2.0 compatible drive
- 11 Kbytes EEPROM programmed via the EEPROM library

WI-FI SUPPORT

Integrated Wi-Fi functionality (not Wi-Fi Shield) is supported in this release.

The Intel® Centrino® Wireless-N 135 adapter that can be plugged into the mini PCIe can be used to add Wi-Fi support. To use the Intel® Centrino® Wireless-N 135 adapter, run the Linux OS from microSD. ■